



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Department of Physics and Astronomy "A. Righi"

Theoretical Physics

Statistical Field Theory

Lectures held by
Prof. Francesco Ravanini

Notes by
Gioele Mancino

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Introduction

1 | Review

In this chapter we will review some basic concepts of statistical mechanics, which will be useful for the rest of the course. We will start with the microcanonical ensemble, which is the simplest ensemble, and then we will move to the canonical ensemble, which is more general and more useful for practical applications. Finally, we will briefly discuss quantum statistical mechanics, which is a generalization of classical statistical mechanics to quantum systems.

1.1 | Equal A Priori Probabilities

We begin recalling the fundamental principles of statistical mechanics, starting from a classical isolated system of N particles, with total energy E and volume V .

The **microstates** of the system are the points in the phase space of the system, which is a $6N$ -dimensional space spanned by the coordinates and momenta of the particles. The **macrostates** of the system are defined by the macroscopic variables, such as energy, volume, and number of particles. The **ensemble** is a collection of microstates that are compatible with a given macrostate. The **probability distribution** of the system is a function that assigns a probability to each microstate in the ensemble.

It is useful to recall the Hamiltonian formalism, which is a powerful tool for describing the dynamics of classical systems. The Hamiltonian H is a function of the coordinates and momenta of the particles, and it encodes the energy of the system. The equations of motion for the system can be derived from the Hamiltonian using Hamilton's equations, which describe how the coordinates and momenta evolve in time. For each particle i we have

$$(q_1, q_2, q_3, p_1, p_2, p_3) \in \mathcal{M},$$

where $\mathcal{M}$ is the phase space of the system, which is a $6N$ -dimensional manifold. The Hamiltonian H is a function that maps points in the phase space to real numbers, and it encodes the energy of the system

$$H = H(q_1, \dots, q_{3N}, p_1, \dots, p_{3N}) : \mathcal{M} \rightarrow \mathbb{R}.$$

The equations of motion for the system can be derived from the Hamiltonian using Hamilton's equations, which describe how the coordinates and momenta evolve in time:

$$\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i}, \\ \dot{p}_i = -\frac{\partial H}{\partial q_i}. \end{cases}$$

We can visualize the phase space of the system as a high-dimensional space, where each point represents a microstate of the system. The energy shell defined by the total energy E is a hypersurface in this phase space, and the trajectories of the system are represented by curves that lie on this hypersurface.

If we suppose that

1. we know positions and momenta of all particles at a given time t , thus we know the microstate of the system at time t ,
2. we have an “infinite-powerful” computer that can solve the equations of motion for the system, thus we can predict the microstate of the system at any time t' given the microstate at time t ,

we should be able to predict the future evolution of the system with perfect accuracy. However, in practice, we do not have access to the microstate of the system, and we do not have an infinite-powerful computer that can solve the equations of motion for the system. Specially if we think about a system with $N \sim N_A = 6 \times 10^{23}$ particles, it is impossible to know the microstate of the system, and it is impossible to solve the equations of motion for the system. We need to introduce

TODO: add a picture of the phase space and the energy shell

approximations and assumptions in order to be able to make predictions about a large collection of particles.

A standard way to introduce statistical mechanics is through the so-called **postulate of equal a priori probabilities**:

Postulate 1.1 (Equal A Priori Probabilities). *Macroscopic observables can be computed defining a **probability distribution** on the phase space of the system, which assigns probabilities to all microstates, and taking the expectation value of the observable with respect to this probability distribution. In an isolated system, all microstates are equally probable, thus the probability distribution is uniform on the energy shell defined by the sub-manifold of the phase space where the energy is equal to E :*

$$\mathcal{M}_E = \{(q_1, \dots, q_{3N}, p_1, \dots, p_{3N}) \in \mathcal{M} : H(q_1, \dots, q_{3N}, p_1, \dots, p_{3N}) = E\}.$$

Let us analyze this postulate in more detail through an exemplification.

Example. Suppose we have a system of N non-interacting particles in a box of volume V , with total energy E . We are interested in computing the average particle velocity $\bar{v}_x$ along some axis x .

We should compute the time dependent function $v_x(t)$ and then take the time average of it:

$$\bar{v}_x(t) = \bar{v}_x(\{q_j(t)\}, \{p_j(t)\}) = \frac{1}{N} \sum_{j=1}^N \frac{v_{x,j}(t)}{m},$$

which is very complicated in general, since we have to solve the equations of motion for the system in order to find the trajectories of the particles in the phase space. However, we can use the postulate of equal a priori probabilities to compute the average particle velocity without solving the equations of motion.

The function $\bar{v}_x(\{q_j\}, \{p_j\})$ is defined on the phase space of the system, and it assigns a value to each microstate of the system. Given the knowledge on some initial microstate, the system will evolve along a curve in the fixed energy shell defined by E :

$$\mathbf{R}(t) = \{q_j(t), p_j(t)\}, \quad \bar{v}_x(t) = \bar{v}_x(\mathbf{R}(t)).$$

We can now assume the system to be at the equilibrium, thus we can take the time average of the function $\bar{v}_x(t) \sim \bar{v}_x$ over a long time interval T as

$$\bar{v}_x = \frac{1}{T} \int_0^T dt \bar{v}_x(\mathbf{R}(t)).$$

Now we can use the postulate of equal a priori probabilities to replace the time average with an average over the energy shell defined by E :

$$\bar{v}_x = \frac{1}{|\mathcal{M}_E|} \int_{\mathcal{M}_E} d\mathcal{M} \bar{v}_x(\{q_j\}, \{p_j\}),$$

where we are using the previous definition of the energy shell $\mathcal{M}_E$ and $|\mathcal{M}_E|$ is the volume of the energy shell in the phase space

$$|\mathcal{M}_E| = \int_{\mathcal{M}_E} d\mathcal{M}.$$

Thus we have computed the average particle velocity without solving the equations of motion for the system, but using only the postulate of equal a priori probabilities: it told us that *averaging over the system trajectory is equivalent to averaging over the whole manifold at fixed energy*.

Let us now analyze the postulate of equal a priori probabilities in more detail. We have to stress some important points about it:

- In typical equilibrium situations, macroscopic quantities (such as $\bar{v}_x$ from the example) usually are constant in time only up to small fluctuations

$$\bar{v}_x(t) = \bar{v}_x + \delta v_x(t).$$

We have previously neglected these fluctuations, but they are actually present in the system, and they can be formally removed by studying the time averages. Indeed we expect those fluctuations to be small and to occur at microscopic time scales, much smaller than the time scales of the macroscopic observables, thus they will not affect the macroscopic properties of the system. Thus measuring (i.e. averaging) over a long time interval T will remove these fluctuations:

$$\frac{1}{\tau} \int_0^\tau dt (\bar{v}_x + \delta v_x(t)) = \frac{1}{\tau} \left(\bar{v}_x \tau + \int_0^\tau dt \delta v_x(t) \right) = \bar{v}_x,$$

where we have used the fact that the time average of the fluctuations is zero.

- The traditional definition of the postulate of equal a priori probabilities is simpler than the one we have given, since it is usually stated as *in an isolated system, all microstates are equally probable*, without specifying that the microstates are the points in the energy shell defined by E . We have given a more modern and precise definition of the postulate, which is more useful for our purposes, since it allows us to compute the average of macroscopic observables as an average over the energy shell in the phase space.
- In the study of dynamical systems, it is possible to mathematically formalize the problem of proving the postulate of equal a priori probabilities, and to find conditions under which it holds. This is a very active field of research, called **ergodic theory**, and in some cases one can prove its validity, while in other cases one can find counterexamples where it does not hold. There exist complicated systems for which the postulate of equal a priori probabilities does not hold, but these systems are not relevant for our purposes, thus we will assume the validity of the postulate for the rest of the course, and we will use it as a fundamental principle.
- We have assumed above that the system is at equilibrium, thus we can take the time average of the macroscopic observable over a long time interval T . If the system is not at equilibrium, then the time average of the macroscopic observable will not be constant in time, and it will depend on the initial conditions of the system. Studying how a system reaches the equilibrium (or even defining it rigorously) is non-trivial, and it is a very active field of research, called **non-equilibrium statistical mechanics** or **thermalization**. Typically the approach to equilibrium is obtained via a weak coupling among the system and the environment, which allows the system to exchange energy and information with the environment, so that the system is not completely isolated.

With the above discussion at hand, it is useful to define the concept of **typical behaviour**. Given a function defined on the phase space (an observable) $f(\{q_j, p_j\})$ associated with some macroscopic

quantity, we can define a **typical point** $\mathbf{R} = \{q_j, p_j\}$ as the point in which the function f takes a value that is close to the average value of the function over the energy shell defined by E :

$$f(\mathbf{R}) = \frac{1}{|\mathcal{M}_E|} \int_{\mathcal{M}_E} d\mathcal{M} f(\{q_j\}, \{p_j\}) + \delta f,$$

where δf is a small fluctuation. The postulate of equal a priori probabilities implies that the typical behaviour of the system is close to the average behaviour of the system, thus we can say that the typical point in the phase space of the system is close to the average value of the function f over the energy shell defined by E .

1.2 | Microcanonical Ensemble

The postulate of equal a priori probabilities defines the microcanonical ensemble (in general an ensemble is a collection of all possible microstates that are consistent with a given set of macroscopic constraints), which is the ensemble of microstates that are compatible with a given macrostate defined by the energy E , volume V , and number of particles N . The probability distribution of the system in the microcanonical ensemble is uniform on the energy shell defined by E , thus we can write it as

$$\rho_{mc}(\{q_j\}, \{p_j\}) = \begin{cases} \frac{1}{|\Gamma_{\Delta}(E)|}, & \text{if } E \leq H(\{q_j\}, \{p_j\}) \leq E + \Delta, \\ 0, & \text{otherwise,} \end{cases} \quad (1.2.1)$$

where Δ is a small energy interval that defines the width of the energy shell, and $|\Gamma_{\Delta}(E)|$ is the volume of the energy shell in the phase space, which can be computed by considering the normalization condition for the probability distribution

$$\int \rho_{mc} d^{3N}q d^{3N}p = 1 \implies \rho_{mc} = \frac{1}{\int_{\Gamma_{\Delta}(E)} d^{3N}q d^{3N}p} = \frac{1}{|\Gamma_{\Delta}(E)|}.$$

Given a function f defined on the phase space of the system, we can compute its **ensemble average**, denoted by $\langle f \rangle_{mc}$ or $\mathbb{E}_{mc}[f]$, as the average of the function over the energy shell defined by E with respect to the probability distribution of the microcanonical ensemble:

$$\langle f \rangle_{mc} = \int f(\{q_j\}, \{p_j\}) \rho_{mc}(\{q_j\}, \{p_j\}) d^{3N}q d^{3N}p. \quad (1.2.2)$$

A *consistency condition* for the microcanonical ensemble (or any other ensemble) is that the statistical fluctuations of macroscopic observables are small in the thermodynamic limit $N \rightarrow \infty$. This means that the variance of a macroscopic observable f should vanish in the thermodynamic limit, thus we can write

$$\sigma_f^2 = \frac{\langle f^2 \rangle_{mc} - \langle f \rangle_{mc}^2}{\langle f^2 \rangle_{mc}} \ll 1 \quad \text{for } N \rightarrow \infty.$$

Now we can use these conditions to further analyze this ensemble; in order to do so let us introduce the entropy S of the system, which can be used to derive the thermodynamic properties of the system, and to find the most probable configuration of the system.

1.2.1 | Entropy

As seen in previous courses, in this framework we are outlining a microscopic basis for the thermodynamic properties of the system, thus we need to introduce a microscopic definition of the entropy S of the system, which is a measure of the number of microstates available to the system at a given energy E . The **Boltzmann formula** for the entropy is given by

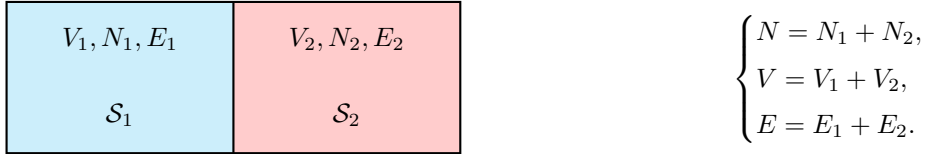
$$S(E) = k_B \log |\Gamma_{\Delta}(E)|, \quad (1.2.3)$$

where k_B is the Boltzmann constant, and $|\Gamma_{\Delta}(E)|$ is the volume of the energy shell in the phase space. The Boltzmann formula for the entropy tells us that the entropy of the system is proportional to the logarithm of the number of microstates available to the system at a given energy E . This is a fundamental result in statistical mechanics, and it allows us to derive the thermodynamic properties of the system from its microscopic properties.

We will not derive the Boltzmann formula for the entropy, but we will use it as a fundamental principle to derive the thermodynamic properties of the system. For example, we can derive a definition of temperature T of the system from the entropy S .

1.2.2 | Temperature

Let us consider two systems $\mathcal{S}_1$ and $\mathcal{S}_2$ in thermal contact such that they can exchange energy, but they are isolated from the rest of the universe. The total system is isolated, as well as the subsystems, thus we can apply the microcanonical ensemble to them.



Thermal contact means that the two systems can exchange energy, thus the total energy E is fixed, but the energy of each subsystem E_1 and E_2 can fluctuate. The total number of particles N and the total volume V are also fixed, since there is no exchange of matter or volume between the two systems.

The total energy can now be computed from the Hamiltonian of the system, sum of the subsystems' Hamiltonians plus an interaction term

$$H(\{q_j\}, \{p_j\}) = H_1(\{q_{1j}\}, \{p_{1j}\}) + H_2(\{q_{2j}\}, \{p_{2j}\}) + \delta H_{12}(\{q_{1j}\}, \{p_{1j}\}, \{q_{2j}\}, \{p_{2j}\}),$$

where the last term accounts for the interaction between the two systems, which we will neglect in the following. The energy of the total system is fixed, but the energy of each subsystem can fluctuate. The probability of finding the system in a state with energy E_1 is proportional to the number of states available to the second system with energy $E_2 = E - E_1$. If we fix an energy shell of width Δ , we can write

$$E_k^{(1)} = k\Delta, \quad E_k^{(2)} = E - E_k^{(1)} = E - k\Delta,$$

and with this “discretization” we can count the number of states available to the total system at a fixed energy E as

$$\Gamma(E) = \sum_{k=0}^{E/\Delta} \Gamma_1(E_k^{(1)}) \Gamma_2(E - E_k^{(1)}),$$

where $\Gamma_k(E) = \int_{E < H_k < E + \Delta} d\{q_{kf}\} d\{p_{kf}\}$ is the number of states available to the k -th subsystem at energy E .

Thus we are saying that *in order to count the number of microstates at energy E we have to multiply the number of microstates of the subsystem $\mathcal{S}_1$ with energy ϵ_1 to the number of microstates of $\mathcal{S}_2$ at energy $\epsilon_2 = E - \epsilon_1$, then sum over all possible values of ϵ_1 .*

The **most probable configuration** of the system is the one that maximizes the number of states available to the total system, which is equivalent to **maximizing the entropy** of the total system. Thus we can compute the entropy of the total system as

$$S(E) = k_B \log \left[\sum_{k=0}^{E/\Delta} \Gamma_1(E_k^{(1)}) \Gamma_2(E - E_k^{(1)}) \right].$$

Now, in order to maximize it, we can take the derivative of the entropy with respect to E_1 and set it to zero: let us denote $\bar{E}_1$ the energy of the subsystem S_1 that maximizes the entropy, then $\bar{E}_2 = E - \bar{E}_1$ is the energy of the subsystem S_2 that maximizes the entropy. We have

$$\Gamma(\bar{E}^{(1)})\Gamma(\bar{E}^{(2)}) \leq \Gamma(E) \leq \frac{E}{\Delta}\Gamma(\bar{E}^{(1)})\Gamma(\bar{E}^{(2)}),$$

where the first inequality comes from the fact that $\Gamma(E)$ is a sum over all the values of E_1 , thus it is greater than or equal to the single term with $E_1 = \bar{E}_1$, while the second inequality comes from the fact that there are at most E/Δ terms in the sum, thus we can upper bound it by multiplying the maximum term by the number of terms. Taking the logarithm of these inequalities we arrive at

$$S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2) \leq S(E, V) \leq S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2) + k_B \log \left(\frac{E}{\Delta} \right),$$

where both inequalities are saturated in the thermodynamic limit $N \rightarrow \infty$: entropy and energy are extensive quantities, thus $S(E, V) \sim N$ and $E \sim N$, so the last term in the upper bound is negligible in the thermodynamic limit

$$k_B \log \left(\frac{E}{\Delta} \right) \sim k_B \log(N) \ll N \sim S(E, V).$$

Thus we can write, in the thermodynamic limit, the entropy of the total system as the sum of the entropies of the subsystems, computed at the most probable values of the energy of the subsystems:

$$S(E, V) = S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2).$$

The first consequence of this result is that the **entropy is an extensive quantity**, since it scales with the number of particles N in the system. The second consequence is that the *most probable configuration of the system is the one that maximizes the entropy*, which is equivalent to maximizing the number of states available to the total system. Thus we can write

$$\Gamma(E) = \Gamma_1(\bar{E}^{(1)})\Gamma_2(\bar{E}^{(2)}).$$

Therefore terms with different values of E_1 are negligible compared to the term with $E_1 = \bar{E}_1$, which is the one that maximizes the number of states available to the total system. This is a consequence of the fact that the number of states available to the total system grows exponentially with the number of particles N , thus even a small difference in energy between different configurations can lead to a huge difference in the number of states available to the total system. $\bar{E}^{(1)}$ and $\bar{E}^{(2)}$ are the most probable values of the energy of the subsystems, which are the ones that maximize the entropy of the total system: the *equilibrium energy values*. Mathematically, we can obtain $\bar{E}^{(1)}$ from the saddle-point condition:

$$\frac{\partial}{\partial \epsilon} [\Gamma_1(\epsilon)\Gamma_2(E - \epsilon)]_{\epsilon=\bar{E}^{(1)}} = 0,$$

which, expanding the derivative, can be written as

$$\left[\frac{\partial}{\partial \epsilon} \Gamma_1(\epsilon) \right]_{\epsilon=\bar{E}^{(1)}} \Gamma_2(E - \epsilon) + \Gamma_1(\epsilon) \left[-\frac{\partial}{\partial \epsilon} \Gamma_2(E - \epsilon) \right]_{\epsilon=\bar{E}^{(2)}} = 0,$$

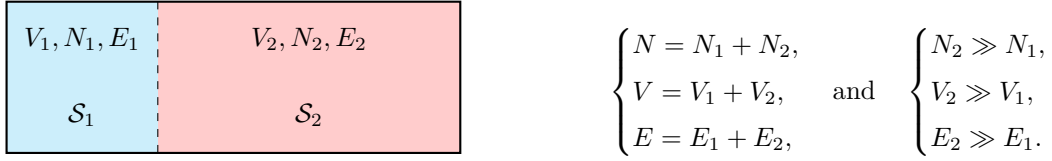
which now, dividing by $\Gamma_1(\bar{E}^{(1)})\Gamma_2(\bar{E}^{(2)})$ (in order to build the logarithmic derivative), and using the definition of entropy we arrive at

$$\left. \frac{\partial}{\partial \epsilon} S_1(\epsilon) \right|_{\epsilon=\bar{E}^{(1)}} = \left. \frac{\partial}{\partial \epsilon} S_2(\epsilon) \right|_{\epsilon=\bar{E}^{(2)}}.$$

This is a fundamental result, which tells us that at equilibrium the derivative of the entropy with respect to the energy is the same for both subsystems. This allows us to define a thermodynamic quantity which is equal in systems in thermal equilibrium, called **temperature** T (recovering the first principle of thermodynamics). The temperature is defined as the inverse of the derivative of the entropy with respect to the energy, thus we can write

$$\frac{1}{T} = \frac{\partial S}{\partial E}(E, V).$$

1.3 | Canonical Ensemble



The second system S_2 is called the *heat bath*, and it is so large that it can exchange energy with the first system S_1 without changing its thermodynamic properties. The total system is isolated, thus we can apply the microcanonical ensemble to it. The probability of finding the system in a state with energy E_1 is proportional to the number of states available to the second system with energy $E_2 = E - E_1$, which is given by $\Omega_2(E - E_1)$.

Technically speaking, we want to find the probability distribution of the subsystem S_1 when it is in contact with a heat bath S_2 , knowing the total probability distribution: the total system is isolated, thus we can apply the microcanonical ensemble to it, and we know that it will be evolving with a trajectory in the phase space that is uniformly distributed on the energy shell defined by the total energy E . This means that if we analyze the evolution of the subsystem S_1 in its phase space, we will find that it will not be limited to fixed energy states, but it will be able to explore all the energy states available to it, with a probability that is proportional to the number of states available to the heat bath S_2 at the corresponding energy. we are interested in finding the **induced probability distribution** of the subsystem S_1 .

Let us visualize the phase space of the total system as a high-dimensional space, where each point represents a microstate of the total system. The energy shell defined by the total energy E is a hypersurface in this phase space, and the trajectory of the total system is uniformly distributed on this hypersurface. The subsystem S_1 can be thought of as a projection of the total system onto a lower-dimensional subspace, where each point represents a microstate of the subsystem S_1 . The probability distribution of the subsystem S_1 is then given by the projection of the uniform distribution on the energy shell onto this lower-dimensional subspace. Different probabilities correspond to different energy states of the subsystem S_1 , and the probability of finding the subsystem S_1 in a state with energy E_1 is proportional to the number of states available to the heat bath S_2 at energy $E - E_1$.

The idea is to fix a point in the phase space of the subsystem S_1 with energy E_1 , and then count the number of points in the phase space of the heat bath S_2 that are compatible with this point, i.e. that have energy $E - E_1$. This will give us the number of states available to the heat bath S_2 at energy $E - E_1$, which is proportional to the probability of finding the subsystem S_1 in a state with energy E_1 .

So we fix an energy ϵ for the subsystem S_1 , and we want to find the number of states available to the heat bath S_2 at energy $E - \epsilon$. *The number of points in the phase space of S that corresponds to P is equal to the number of points in the phase space of S_2 that corresponds to $E - \epsilon$.* Thus we can write

$$\rho(\{q_f\}, \{p_f\}) \propto \frac{\Gamma_2(E - \epsilon)}{\Gamma(E)}.$$

We know that the entropy of the heat bath S_2 is related to the number of states available to it by

the Boltzmann formula (we are using $k_B = 1$ for simplicity), such that

$$\Gamma_2(E - \epsilon) = e^{S_2(E - \epsilon)} \sim e^{S_2(E) - \epsilon \frac{\partial S_2}{\partial E}},$$

where we have used the fact that the entropy is an extensive quantity, thus we can write it as a function of the energy per particle, and we have expanded it around the total energy E of the heat bath S_2 . We can now recall the temperature as a thermodynamic quantity defined as the inverse of the derivative of the entropy with respect to the energy, while reinserting the Boltzmann constant k_B for clarity, we have

$$\rho(\{q_f\}, \{p_f\}) \propto e^{-\beta \epsilon} = e^{-\frac{\epsilon}{k_B T}},$$

where $\beta = \frac{1}{k_B T}$ is the inverse temperature. Thus we have derived the Boltzmann distribution for the subsystem S_1 in contact with a heat bath S_2 . The probability of finding the subsystem S_1 in a state with energy ϵ is proportional to the exponential of the negative of the energy divided by the temperature of the heat bath S_2 . This is the canonical ensemble, where the subsystem S_1 is in thermal equilibrium with a heat bath S_2 at temperature T .

We can now define the partition function Z as the normalization constant for the probability distribution of the subsystem S_1 , such that

$$Z = \int \frac{d^{3N}q d^{3N}p}{h^{3N} N!} e^{-\beta H(\{q\}, \{p\})},$$

which can be used as a generating function for the thermodynamic properties of the subsystem S_1 . For example, the free energy F can be computed from the partition function as

$$F = -k_B T \log Z = -\frac{1}{\beta} \log Z,$$

and the internal energy U can be computed as

$$U = -\frac{\partial}{\partial \beta} \log Z,$$

while the entropy S can be computed as

$$S = -\frac{\partial F}{\partial T} = k_B \beta^2 \frac{\partial F}{\partial \beta}.$$

Discrete System

A discrete system is a system that can only take a finite number of states, such as a spin system or a lattice gas. In this case, let us imagine a collection of N particles that can only occupy a finite number of states, let us imagine two values of spin, up and down, thus we have 2^N possible configurations of the system. The phase space of the system is now a discrete set of points, configurations (a set of sets) $\{c_j\}_{j=1}^N$, and the probability distribution of the system is defined on this discrete set of points. The partition function can be written as a sum over all possible configurations of the system, such that

$$Z = \sum_{\{c_j\}} e^{-\beta H(\{c_j\})}.$$

1.4 | Quantum Statistical Mechanics

Considering a general quantum mechanical system, we can study it with different tools, such as the hamiltonian operator, the density matrix, the partition function, etc. The Hamiltonian operator H is a self-adjoint operator that acts on the Hilbert space of the system, and it encodes the energy of the system. The density matrix ρ is a positive semi-definite operator that acts on the Hilbert space of the system, and it encodes the state of the system. The partition function Z is a scalar quantity that encodes the thermodynamic properties of the system.

$$\begin{aligned}
 \hat{H} |n\rangle &= E_n |n\rangle, \\
 |\psi\rangle &\in \mathcal{H}, \\
 \{|E_n\rangle\}_{n=1}^D &\implies |\psi\rangle = \sum_{n=1}^D c_n |E_n\rangle, \\
 \langle\psi| \hat{A} |\psi\rangle &= \sum_{n,m} c_n^* c_m \langle E_n | \hat{A} | E_m \rangle, \\
 |\psi(t)\rangle &= \sum_n c_n(t) |E_n\rangle, \quad c_n(t) = c_n e^{-i \frac{E_n}{\hbar} t} \\
 \langle\psi(t)| \hat{A} |\psi(t)\rangle &= \sum_{n,m} c_n^* c_m e^{i \frac{E_n - E_m}{\hbar} t} \langle E_n | \hat{A} | E_m \rangle, \\
 \implies &= \frac{1}{T} \int_0^T dt \langle\psi(t)| \hat{A} |\psi(t)\rangle = \sum_n |c_n|^2 \langle E_n | \hat{A} | E_n \rangle.
 \end{aligned}$$

Let us report some postulates of quantum statistical mechanics:

- Postulate of equal a priori probabilities: in an isolated system, all microstates are equally probable,

$$|\bar{c}_n|^2 = \begin{cases} \frac{1}{D}, & \text{if } E_n \in [E, E + \Delta], \\ 0, & \text{otherwise.} \end{cases}$$

- postulate of random phases: the phases of the coefficients c_n are random variables uniformly distributed in the interval $[0, 2\pi]$, thus

$$\frac{1}{T} \int_0^T dt c_n^*(t) c_m(t) = \frac{1}{T} \int_0^T dt c_n^* c_m e^{i \frac{E_n - E_m}{\hbar} t} = \begin{cases} |c_n|^2, & \text{if } n = m, \\ 0, & \text{otherwise.} \end{cases}$$

The time average of an observable $\hat{A}$ is given by

$$\overline{\langle\psi(t)| \hat{A} |\psi(t)\rangle} = \frac{1}{T} \int_0^T dt \langle\psi(t)| \hat{A} |\psi(t)\rangle = \sum_n |c_n|^2 \langle E_n | \hat{A} | E_n \rangle.$$

[...]

We can define the density matrix ρ as

$$\rho = \frac{1}{Z} \sum_{E \leq E_n \leq E + \Delta} |E_n\rangle \langle E_n|,$$

which is a positive semi-definite operator that encodes the state of the system. The expectation value of an observable $\hat{A}$ can be then computed as

$$\langle \hat{A} \rangle = \text{Tr}(\rho \hat{A}).$$

We have to note that dealing with quantum statistical mechanics is more complicated than dealing with classical statistical mechanics, since we have to deal with operators instead of functions, and we have to deal with the non-commutativity of operators. However, the general principles of statistical mechanics still apply, and we can still define ensembles, partition functions, and thermodynamic quantities in the quantum case. The main difference is that we have to use the language of operators and Hilbert spaces instead of the language of functions and phase space.

Defining an ensemble in quantum statistical mechanics automatically defines a density matrix, which encodes the state of the system, and from which we can compute the expectation values of observables and the thermodynamic properties of the system. We are governing our set of eigenstates with a probability distribution, which is encoded in the density matrix, and we can compute the expectation values of observables and the thermodynamic properties of the system from this density matrix.

For a quantum system at thermal equilibrium, we have the following identification of the thermodynamic free energy F with the partition function Z :

$$F = -k_B T \log Z,$$

which is the same relation that we have in classical statistical mechanics. These postulates are like operational tools: justify them as they are can be really difficult and it is an actual field of research with a lot of literature, but they are very useful for computing the thermodynamic properties of quantum systems at thermal equilibrium. For us, justifying why a precise postulate worked in the actual setting will be enough.

2 | Phase Transitions

Phase Transitions are fundamental phenomena in statistical physics, where a system undergoes a qualitative change in its macroscopic properties as external parameters (like temperature or pressure) are varied. These transitions can be classified into different types based on their characteristics and the nature of the change in the system. Let us start from the definition of a phase of a many-particle system, which is crucial for understanding phase transitions.

Definition 2.1 (Phase of Many-Particle System). A phase of a many-particle system is a region in the parameter space (e.g., temperature, pressure) where the system exhibits distinct physical properties and behavior: it is a “type” of collective state of matter. They are defined in terms of macroscopic properties such as density, magnetization, or symmetry.

A phase transition occurs when a system undergoes an *abrupt change from one phase to another* (states) *as we smoothly vary an external parameter*. Often PTs are accompanied by a discontinuity or singularity in some thermodynamic quantity, which serves as an order parameter to characterize the transition.

A phase transition is always a result of a competition between “order” and “disorder” in the system. For example, in a ferromagnetic material, the ordered phase corresponds to a state where the magnetic moments are aligned, while the disordered phase corresponds to a state where the magnetic moments are randomly oriented. The transition between these two phases can be driven by temperature, where thermal fluctuations disrupt the order at high temperatures.

We are now going to define PTs in a mathematical framework, which will allow us to classify them into different types. To start we have to highlight how PTs are related to singularities in the free energy of the system. The free energy is a thermodynamic potential that encodes the equilibrium properties of a system, and its behavior near a phase transition can reveal important information about the nature of the transition:

$$F = -\frac{1}{\beta} \ln Z = U - TS,$$

where Z is the partition function of the system and $\beta = 1/(k_B T)$ is the inverse temperature. A phase transition is typically associated with a non-analytic behavior of the free energy as a function of the external parameters, such as temperature or pressure. Thus PTs are defined as singularities in the thermodynamic limit of the partition function Z or equivalently of the free energy F .

Definition 2.2 (Phase Transition). A phase transition is a non-analytic behavior of the free energy F as a function of external parameters (e.g., temperature, pressure) in the thermodynamic limit. This non-analyticity manifests as a discontinuity or singularity in some thermodynamic quantity, which serves as an order parameter to characterize the transition.

Example (Finite system and phase transitions). If we consider a finite set of spins, the partition function Z is a finite sum of exponentials, which is an analytic function of the parameters:

$$Z = \sum_{\{\text{conf}\}} e^{-\beta E[\text{conf}]}.$$

Therefore, in a finite system, there can be no true phase transition, as the free energy will always be analytic. However, as we take the thermodynamic limit (i.e., let the number of spins go to infinity), the partition function can develop singularities, leading to non-analytic behavior in the free energy and thus allowing for phase transitions to occur.

In physics we never encounter real infinite systems, but we can approximate them by considering large systems and taking the thermodynamic limit. This is a common approach in statistical mechanics to study phase transitions, as it allows us to capture the essential features of the transition while avoiding the complications that arise from finite-size effects.

2.1 | Classification of Phase Transitions

Phase transitions can be classified into different types based on their characteristics and the nature of the change in the system. One of the first classifications was proposed by Paul Ehrenfest, who categorized phase transitions based on the order of the derivative of the free energy that exhibits a discontinuity at the transition point. Today, we have a more refined classification, which is based on the concept of order parameters and symmetry breaking.

Phase transitions can be classified depending on the type of singularity in the free energy. We can distinguish between first-order and continuous (second-order) phase transitions:

- **First-order phase transitions** are characterized by a discontinuity in the first derivative of the free energy with respect to some external parameter (e.g., temperature). This means that there is a **latent heat** associated with the transition, and the order parameter changes discontinuously.¹ Examples include the liquid-gas transition and the melting of a solid.
- **Continuous (second-order) phase transitions** are characterized by a continuous first derivative of the free energy, but a discontinuity in the second or higher derivatives. In this case, there is no latent heat, and the order parameter changes continuously. Examples include the ferromagnetic transition in the Ising model and the superconducting transition in certain materials. These are the most interesting PTs and will drive most of our discussion in the following chapters.

The latter group of PTs is also called **critical phenomena**, and they are associated with the emergence of long-range correlations and *scale invariance* at the critical point. The study of critical phenomena has led to the development of powerful theoretical tools, such as the *renormalization group*, which allows us to understand the universal behavior of systems near criticality. Another important phenomenon associated with continuous phase transitions is the concept of **spontaneous symmetry breaking**. In many cases, the ordered phase of a system exhibits a lower symmetry than the disordered phase. This spontaneous symmetry breaking is a key feature of many continuous phase transitions and plays a crucial role in determining the properties of the system near criticality.

2.1.1 | Phase Diagram of Water

Looking at the phase diagram of water, we can see that there are three main phases: solid (ice), liquid (water), and gas (vapor). The phase boundaries between these phases are represented by lines in the pressure-temperature (p-T) diagram. The point where the liquid-gas phase boundary ends is called the **critical point**, which is a second-order phase transition. At this point, the properties of the liquid and gas phases become indistinguishable, and the system exhibits critical phenomena such as large fluctuations and long-range correlations.

TODO: Draw p-T diagram of water, showing the different phases and the critical point.

Moving along the phase boundaries, we can observe first-order phase transitions, where the system undergoes a discontinuous change in density and other properties. For example, when water freezes into ice or boils into vapor, there is a latent heat associated with the transition, and the order parameter (density) changes discontinuously. The first derivative of the free energy with respect

¹This happens because we can express order parameters as derivatives of the free energy, which is continuous but has discontinuous first derivative.

to temperature (which is related to the entropy) and pressure (which is related to the volume) will show a discontinuity at these transitions, confirming their first-order nature.

TODO: Draw the phase diagram of water, $p - v = V/N$

Decreasing the pressure at a fixed temperature can also lead to phase transitions. For example, if we decrease the pressure of liquid water at a constant temperature, we can reach the point where it transitions into vapor. This is another example of a first-order phase transition, as there is a discontinuity in the density and other properties of the system. The phase diagram of water in the pressure-volume ($p-v$) plane also shows the different phases and the transitions between them, with the critical point marking the end of the liquid-gas phase boundary.

If we are below the critical temperature, at a fixed T we can observe all three phases by varying the pressure: this line forms a mesoscopic domain where we can observe the coexistence of phases and the transitions between them. This is a rich area of study in statistical mechanics, as it allows us to explore the behavior of systems near criticality and understand the underlying mechanisms driving phase transitions. It is named **coexistence line**.

TODO: draw the $p-v-T$ diagram of water.

The full phase diagram of water in the pressure-volume-temperature ($p-v-T$) space provides a comprehensive view of the different phases and the transitions between them. It shows how the phases of water change as we vary both temperature and pressure, with the critical point marking the end of the liquid-gas phase boundary. This diagram is essential for understanding the behavior of water under different conditions and is a fundamental tool in thermodynamics and statistical mechanics.

Transitions may be direction dependent in the phase space... At the critical point, the system exhibits critical phenomena such as large fluctuations and long-range correlations. The properties of the system near the critical point can be described by critical exponents, which characterize how various physical quantities diverge or vanish as the critical point is approached. These critical exponents are universal, meaning that they depend only on the dimensionality of the system and the symmetry of the order parameter, rather than on the specific details of the microscopic interactions. This universality is a key feature of critical phenomena and allows us to classify different systems into universality classes based on their critical behavior.

Phase transitions are well defined when described with a curve in the parameter space, but we can also have transitions that occur at a single point, such as the critical point in the phase diagram of water. In this case, the transition is not associated with a curve but rather with a specific set of conditions (temperature and pressure) where the properties of the system change dramatically.

We are only going to discuss PTs which occurs with a smooth variation of the parameters, but there are also PTs that can occur with a sudden change in the parameters, such as a quench or a rapid cooling. These non-equilibrium phase transitions can lead to interesting phenomena such as the formation of topological defects and the emergence of new phases that are not present in equilibrium conditions. The study of non-equilibrium phase transitions is an active area of research in statistical mechanics and condensed matter physics.

2.1.2 | Magnetic System

In this setting, we are considering a universal magnetic system, which can be described by a Hamiltonian that includes interactions between magnetic moments (spins) and an external magnetic field. The behavior of the system can be analyzed in terms of its phase transitions, which are characterized by changes in the order parameter (magnetization) as we vary the temperature

and the external magnetic field.

We have few control parameters over the system: the temperature T and the external magnetic field h . The phase diagram of the magnetic system can be represented in the $T - h$ plane, where we can identify different phases based on the behavior of the magnetization.

TODO: Add a figure showing the phase diagram of a magnetic system in the T-h plane.

- At high temperatures and low magnetic fields, the system is in a disordered phase where the magnetization is zero: **paramagnetic phase**.
- At low temperatures and high magnetic fields, the system is in an ordered phase where the magnetization is non-zero: **ferromagnetic phase**.
- The transition between these two phases can be either first-order or second-order, depending on the specific interactions in the system. In many cases, there is a critical point at which the nature of the transition changes from first-order to second-order.
- At the critical point, the system exhibits critical phenomena such as large fluctuations in magnetization and long-range correlations. The properties of the system near the critical point can be described by critical exponents, which characterize how various physical quantities diverge or vanish as the critical point is approached.

Introducing the **magnetization** as the order parameter of the system, we can analyze how it changes across the phase transitions.

Definition 2.3 (Order Parameter). An order parameter is a quantity that serves as a measure of the degree of order in a system and can be used to characterize different phases. It typically takes on a non-zero value in the ordered phase and vanishes in the disordered phase. It is expressed generally as

$$m(T) = \frac{1}{V} \lim_{h \rightarrow 0} \langle M(h, T) \rangle,$$

where the notation is deliberately describing the case of magnetization, but we can interpret more generally as the order parameter of a system, which can be defined in terms of the expectation value of some observable that characterizes the order in the system: thus we interpret in general as

$$\phi(T) = \frac{1}{V} \lim_{h \rightarrow 0} \langle O(h, T) \rangle,$$

where $O(h, T)$ is an observable that characterizes the order in the system, and V is the volume of the system, h is an external field that couples to the order parameter, and the limit $h \rightarrow 0$ ensures that we are measuring the spontaneous magnetization (or order parameter) in the absence of an external field.

In the paramagnetic phase, the magnetization is zero, while in the ferromagnetic phase, it takes on a non-zero value. The behavior of the magnetization near the critical point can be described by a power law, with a critical exponent that characterizes how the magnetization vanishes as we approach the critical point from the ordered phase.

This is equivalent to the jump of the volume in the liquid-gas transition (p-v water diagram), which is the order parameter of that system. The magnetization serves as a measure of the degree of order in the magnetic system, and it is associated to a first derivative of the free energy, which is discontinuous at a first-order phase transition and continuous at a second-order phase transition. At any temperature below the critical temperature, we can observe a first-order phase transition

TODO: Add a figure showing the behavior of the magnetization as a function of temperature near the critical point.

by varying the external magnetic field. As we increase the magnetic field from zero, the system transitions from the paramagnetic phase to the ferromagnetic phase, with a discontinuous change in magnetization. This is a first-order phase transition, as there is a latent heat associated with the transition and the order parameter (magnetization) changes discontinuously.

The critical temperature separates two phases with different symmetries: the paramagnetic phase has a higher symmetry (rotational symmetry) than the ferromagnetic phase (which has a lower symmetry due to the alignment of spins). This is an example of spontaneous symmetry breaking, where the ordered phase exhibits a lower symmetry than the disordered phase. The study of this magnetic system and its phase transitions provides valuable insights into the behavior of many other systems that exhibit similar phenomena, such as superconductors and liquid crystals.

2.1.3 | Critical Behaviour

When the order parameter approaches zero continuously at the critical point, we can describe its behavior using a power law:

$$m(T) = \begin{cases} 0 & \text{for } T > T_c, \\ (T_c - T)^\beta & \text{for } T < T_c, \end{cases}$$

where β is the critical exponent that characterizes how the magnetization vanishes as we approach the critical point from the ordered phase. The value of β depends on the universality class of the system, which is determined by factors such as the dimensionality of the system and the symmetry of the order parameter. For example, in the three-dimensional Ising model, which is a prototypical model for ferromagnetic phase transitions, the critical exponent β is approximately 0.326.

The surprising thing is that if we compute the same exponent for a completely different system, such as the liquid-gas transition, using some difference in densities as the order parameter, we find the same value of β :

$$|\rho_l - \rho_g| \sim (T_c - T)^\beta,$$

where ρ_l and ρ_g are the densities of the liquid and gas phases, respectively, and the critical exponent β is exactly the same as in the magnetic system, despite the fact that these systems have different microscopic details and interactions.

This is a manifestation of the **universality of critical phenomena**, which states that systems with different microscopic details can exhibit the same critical behavior near the phase transition, as long as they belong to the same universality class. This universality allows us to classify different systems based on their critical exponents and understand their behavior near criticality without needing to know the specific details of their interactions.

2.1.4 | Critical Exponents and Universality Classes

The critical exponents are a set of numbers that characterize the behavior of physical quantities near the critical point. They describe how various properties of the system diverge or vanish as the critical point is approached. Some of the most common critical exponents include:

- β : describes the behavior of the **order parameter** *as a function of temperature* near the critical point, $m(T) \sim (T_c - T)^\beta$ for $T < T_c$. We have already discussed this exponent in the context of the magnetization and the liquid-gas transition, where it characterizes how the order parameter vanishes as we approach the critical point from the ordered phase.

- α : describes the divergence of the specific heat (heat capacity) at constant volume:

$$C_V \sim |T - T_c|^{-\alpha},$$

and the heat capacity is defined as the second derivative of the free energy with respect to temperature:

$$C_V = \frac{\partial U}{\partial T} = [\dots] = -T \frac{\partial^2 F}{\partial T^2}.$$

TODO: complete calculations missing.

- δ : describes the behavior of the **order parameter** *as a function of the external field* at the critical temperature,

$$m(h) \sim h^{1/\delta} \quad \text{at } T = T_c,$$

where h is the external magnetic field.

- γ : describes the divergence of the **susceptibility** *as a function of temperature*:

$$\chi \sim |T - T_c|^{-\gamma},$$

where the susceptibility χ is defined as the derivative of the order parameter with respect to the external field:

$$\chi = \frac{\partial m}{\partial h}, \quad \chi(t \rightarrow 0^\pm) \sim |t|^{-\gamma_\pm},$$

with $t = (T - T_c)/T_c$ being the reduced temperature. The susceptibility measures how responsive the system is to changes in the external field, and its divergence at the critical point indicates that even a small change in the magnetic field can lead to a large change in the magnetization. This is a hallmark of criticality and is associated with the emergence of long-range correlations in the system.

Before continuing we have to introduce the concept of correlation length, which is a measure of how far correlations in the system extend. Near the critical point, the correlation length diverges, indicating that fluctuations occur over increasingly large distances. Let us introduce it through a little model: let us consider a system of spins on a lattice, where each spin s_i can take on values of +1 or -1. The magnetization m is defined as the average spin per site:

$$m = \frac{1}{N} \sum_{i=1}^N \langle s_i \rangle,$$

where the angle brackets denote the thermal average:

$$\langle s_i \rangle = \frac{1}{Z} \sum_{\{s_k\}} s_i e^{-\beta H(\{s_k\}, h)},$$

with Z being the partition function and H the Hamiltonian of the system. The correlation function $G(r)$ is defined as the average product of spins at a distance r :

$$G_{ij}^{(2)} = \langle s_i s_j \rangle = \frac{1}{Z} \sum_{\{s_k\}} s_i s_j e^{-\beta H(\{s_k\}, h)} = G^{(2)}(\mathbf{r}), \quad \text{where } \mathbf{r} = |\mathbf{r}_i - \mathbf{r}_j|.$$

We can now define the connected correlation function, which measures the correlations between fluctuations in the system:

$$G_c^{(2)}(\mathbf{r}) = \langle (s_i - \langle s_i \rangle) (s_j - \langle s_j \rangle) \rangle = \langle s_i \rangle \langle s_j \rangle - \langle s_i \rangle \langle s_j \rangle.$$

The correlation length ξ is defined as the characteristic length scale over which correlations decay in the system. Near the critical point, the correlation function typically decays as a power law:

$$G_c^{(2)}(\mathbf{r}) \sim e^{-r/\xi} \quad \text{for } r \gg \xi,$$

Now we can introduce two more critical exponents that describe the behavior of the correlation function and the correlation length near the critical point:

- ν : describes the divergence of the correlation length, $\xi_{\pm} \sim |T - T_c|^{-\nu_{\pm}}$, with $\nu_{\pm} > 0$.
- η : describes the behavior of the correlation function at the critical point, $G(r) \sim r^{-(d-2+\eta)}$ at $T = T_c$, where d is the dimensionality of the system.

Now we are ready to define and discuss the concept of universality classes. Systems that share the same critical exponents and scaling behavior near the critical point are said to belong to the same universality class. The universality class of a system is determined by factors such as the dimensionality of the system, the symmetry of the order parameter, and the range of interactions between particles.

Definition 2.4 (Universality Class). A universality class is a category of systems that share the same critical exponents and scaling behavior near the critical point, regardless of their microscopic details. Systems that belong to the same universality class exhibit the same critical behavior and can be described by the same effective field theory near the critical point. The universality class of a system is determined by factors such as the dimensionality of the system, the symmetry of the order parameter, and the range of interactions between particles.

Let us now make some observations about the critical behavior of systems near phase transitions, which will be important for our understanding of critical phenomena and the renormalization group.

Remark. *There is a connection between the divergence of the correlation length the response function. The connection relies on the fact that near the critical point, the system exhibits long-range correlations, which means that fluctuations occur over increasingly large distances. This leads to a divergence in the correlation length ξ , which in turn causes the response function (such as susceptibility) to diverge as well.*

The response function measures how the system responds to external perturbations, and its divergence at the critical point indicates that even a small change in the external field can lead to a large change in the order parameter, which is a hallmark of criticality.

Remark. *The **milky appearance** (or critical opalescence) of a system near the critical point is a direct consequence of the divergence of the correlation length ξ (for example in a liquid-gas mixture). As the correlation length increases, fluctuations in the density or order parameter occur over larger and larger distances, leading to the scattering of light and giving rise to a milky or cloudy appearance. This phenomenon is a visual manifestation of the critical behavior of the system and is often observed in fluids near their critical point, such as in the case of water at its critical temperature and pressure.*

Remark. *Second order phase transitions are also called **continuous phase transitions** because the order parameter changes continuously across the transition, and there is no latent heat associated with the transition. As we will see, these PTs are associated with the **spontaneous***

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***symmetry breaking.** In some cases, the symmetry is only **emergent**, meaning that it is not present in the microscopic description of the system but emerges as a result of the collective behavior of the system near the critical point (approximate symmetry which becomes exact only near the critical point). This emergent symmetry can play a crucial role in determining the properties of the system and its critical behavior.*

2.2 | Mean Field Theory

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Mean field theory is a powerful approximation method used to study phase transitions and critical phenomena in many-body systems. The basic idea behind mean field theory is to replace the complex interactions between particles in the system with an average or "mean" field that each particle experiences. This simplification allows us to derive self-consistent equations for the order parameter and analyze the behavior of the system near the critical point.

Let us concentrate on a classical model, the Ising model, which is a prototypical model for studying ferromagnetic phase transitions.

2.2.1 | Ising Model

TODO: insert
images of spins on
a lattice

The Ising model consists of a lattice of spins, where each spin s_i can take on values of $+1$ or -1 . The Hamiltonian of the Ising model is given by:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i,$$

where J is the coupling constant that determines the strength of the interaction between neighboring spins, h is the external magnetic field, and the sum $\langle i, j \rangle$ runs over pairs of **nearest-neighbor** spins. The first term in the Hamiltonian represents the interaction between neighboring spins, which favors alignment (ferromagnetic order) when $J > 0$. The second term represents the coupling of the spins to the external magnetic field, which tends to align the spins in the direction of the field.

Definition 2.5 (Coordination Number). The coordination number z of a lattice is the number of nearest neighbors that each site has. For example, in a two-dimensional square lattice, each site has 4 nearest neighbors, so the coordination number is $z = 4$. In a three-dimensional cubic lattice, each site has 6 nearest neighbors, so the coordination number is $z = 6$. The coordination number plays an important role in determining the behavior of the system and the nature of the phase transition.

Our aim is to study the equilibrium properties (where observables can effectively be calculated through thermal averaging) of the Ising model and understand the nature of the phase transition it exhibits. To do this, we will apply mean field theory to derive self-consistent equations for the magnetization (order parameter) and analyze the behavior of the system near the critical point.

Each configuration of the system is specified by the set of spin values $\{s_i\}_{i=1}^N$, and it is associated to a probability given by the Boltzmann distribution:

$$P(\{s_i\}) = \frac{1}{Z} e^{-\beta H(\{s_i\})},$$

where Z is the **partition function**, defined as the sum over all possible configurations:

$$Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}.$$

Example (Pair of spins). When we have only two spins, the partition function can be calculated explicitly. The Hamiltonian for two spins s_1 and s_2 is given by:

$$H = -Js_1 s_2 - h(s_1 + s_2),$$

where the sum is removed since we only have two spins. We can now compute the energy for each of the four possible configurations of the spins:

- $s_1 = +1, s_2 = +1$: $H = -J - 2h$.
- $s_1 = +1, s_2 = -1$: $H = J$.
- $s_1 = -1, s_2 = +1$: $H = J$.
- $s_1 = -1, s_2 = -1$: $H = -J + 2h$.

The partition function is then given by:

$$Z = e^{\beta J + 2\beta h} + 2e^{-\beta J} + e^{\beta J - 2\beta h}.$$

If we now wanted to compute the probability of finding the system in the configuration $s_1 = +1, s_2 = +1$, we would use the Boltzmann distribution:

$$P(s_1 = +1, s_2 = +1) = \frac{e^{\beta J + 2\beta h}}{Z}.$$

For the magnetization, we can compute the average spin per site:

$$M = \mathbb{E}_c[\sum_i s_i], \quad m = \frac{M}{N} = \frac{1}{N} \mathbb{E}_c[s_1 + s_2],$$

where the summations run over the configuration space.

$$m = \sum_{\langle s_i \rangle} p(\{s_i\}) \left(\frac{1}{N} \sum_i s_i \right),$$

where the sum runs over all possible configurations of the spins. In this case, we can explicitly compute the magnetization by summing over the four configurations.

If we were now to consider a system with $N = 100$, we would have 2^{100} possible configurations for a chain with $z = 2$, which is an astronomically large number. This makes it impossible to compute the partition function and the magnetization by explicitly summing over all configurations. This is where mean field theory comes into play, as it allows us to approximate the behavior of the system without needing to consider every possible configuration.

2.2.2 | Mean Field Approach - Euristic

Let us now apply the mean field approximation to the Ising model. The key idea is to replace the interaction term in the Hamiltonian with an average or "mean" field that each spin experiences. This allows us to derive a self-consistent equation for the magnetization.

It is a *euristic* approach, which is not exact but provides a good approximation, remaining not systematic, not rigorous and hard to fully justify. We will see how it can be derived from a more rigorous approach, which is the **Landau-Ginzburg effective field theory**.

There are many versions of this approach (different implementations of the approximation), but the idea is almost the same: we want to replace some microscopic degrees of freedom with an average field, which is the mean field. The mean field is defined as the average value of the spin at a given site:

Let us rewrite the Hamiltonian of the Ising model

$$H = -J \sum_{\langle s_i, s_j \rangle} s_i s_j - h \sum_i s_i.$$

If we now consider the case in which the coupling constant J is positive ($J > 0$ which corresponds to a ferromagnetic interaction), and a null external magnetic field ($h = 0$), the Hamiltonian simplifies to:

$$H = -J \sum_{\langle s_i, s_j \rangle} s_i s_j.$$

In this case the system exhibits a phase transition at a critical temperature T_c , below which the spins tend to align and the system enters an ordered ferromagnetic phase, while above T_c the spins are disordered and the system is in a paramagnetic phase.

Being interested in the behavior of the magnetization, we can express it as a thermal average in the canonical ensemble:

$$\mathbb{E}_c \left[\frac{1}{N} \sum_i s_i \right] = \frac{1}{N} \sum_{\{s_i\}} \left(\sum_i s_i \right) \frac{e^{-\beta H(\{s_i\})}}{Z},$$

where the sum runs over all possible configurations of the spins. The mean field approximation consists in building a term which encodes the average effect of the neighboring spins on a given spin: it is natural to think about building this average field from the magnetization, so let us analyze it in more detail. The magnetization can assume the following values:

$$M \in [-N, -N+2, -N+4, \dots, +N],$$

where the values change by 2 because each spin can flip from +1 to -1 or vice versa. The magnetization per site is then given by:

$$m = \frac{M}{N} \in \left[-1, -1 + \frac{2}{N}, -1 + \frac{4}{N}, \dots, +1 \right].$$

The average field we are building has to be proportional to the magnetization m , which serves as a measure of the degree of order in the system. Thus we can express the sum over configurations of the spins as a sum over the magnetization M and then a sum over all configurations that correspond to that magnetization:

$$\sum_{\{s_i\}_{i=1}^N} [\dots] = \sum_{M=-N}^N \sum_{\{s_i\}_{i=1}^N} \delta \left(\sum_i s_i - M \right) [\dots],$$

where we have introduced a delta function to enforce the constraint that the sum of the spins equals the magnetization M . This allows us to rewrite the partition function as a sum over the magnetization, but let's understand better the meaning of this delta function through an example.

Example ($N = 2$). If we have only two spins, the possible configurations and their corresponding magnetizations are:

- A: $s_1 = +1, s_2 = +1$: $M = 2$.
- B: $s_1 = +1, s_2 = -1$: $M = 0$.
- C: $s_1 = -1, s_2 = +1$: $M = 0$.
- D: $s_1 = -1, s_2 = -1$: $M = -2$.

The delta function $\delta(\sum_i s_i - M)$ ensures that we only sum over configurations that correspond to a specific magnetization M .

Let us analyze the contribution of each configuration to the partition function: starting from a sum over all configurations, we get

$$Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})} = e^{-\beta H(A)} + e^{-\beta H(B)} + e^{-\beta H(C)} + e^{-\beta H(D)},$$

where $H(A)$, $H(B)$, $H(C)$, and $H(D)$ are the Hamiltonian evaluated at each configuration: considering the case $h = 0$ for simplicity, we have

$$H(A) = -J, \quad H(B) = J, \quad H(C) = J, \quad H(D) = -J.$$

Thus the partition function can be rewritten as

$$Z = 2(e^{\beta J} + e^{-\beta J}).$$

We now want to get the same result using the sum over magnetization, which is given by

$$Z = \sum_{M=-2}^2 \sum_{\{s_i\}} \delta\left(\sum_i s_i - M\right) e^{-\beta H(\{s_i\})} = \sum_{M=-2}^2 e^{-\beta H(M)} \sum_{\{s_i\}} \delta\left(\sum_i s_i - M\right) = 2(e^{\beta J} + e^{-\beta J}),$$

where we have used the fact that the Hamiltonian depends only on the magnetization M and not on the specific configuration of the spins, and that the sum over configurations for each magnetization value is just the number of configurations with that value of magnetization. This example illustrates how the sum over configurations can be rewritten as a sum over magnetization, with the delta function ensuring that we only consider configurations that correspond to a specific magnetization.

Now we can implement the mean field approximation by replacing the interaction term in the Hamiltonian with an average field that each spin experiences. This average field is proportional to the magnetization m , which serves as a measure of the degree of order in the system: in a fixed magnetization sector (m-sector) we can replace the interaction term with an effective field that depends on the magnetization (reinserting the external magnetic field h):

$$H_{mf}(\{s_i\}, m) = -J \sum_{\langle i,j \rangle} s_i m - h \sum_i s_i,$$

where we are summing over single sites (only s_i appears in the sum), but we kept the notation of the sum over nearest neighbors to indicate that the effective field is proportional to the number of nearest neighbors (coordination number z) and the magnetization m . This is the point where we have to accept that this method works even if it is hard to justify: there exist some ad-hoc argument, but it would be meaningless to use those since we will anyway present the Landau-Ginzburg effective field theory in the following sections, which then will make us able to justify this approximation in a systematic way.

Remark. The hamiltonian H_{mf} is independent of the specific configuration of the spins $\{s_i\}$, as long as

$$\sum_i s_i = M,$$

So that we have efficiently replaced the interaction term with an average field that depends only on the magnetization m . This allows us to derive a self-consistent equation for the magnetization and analyze the behavior of the system near the critical point without needing to consider every possible configuration of the spins.

If we continue the computation of the mean field hamiltonian we get

$$H_{mf}(\{s_i\}, m) = -Jm \sum_{\langle i,j \rangle} s_i - h \sum_i s_i = -(Jzm + h) \sum_i s_i = -(Jzm + h)Nm,$$

where we have used the fact that the sum over the spins is equal to the magnetization $M = Nm$ and the sum over the nearest neighbors gives a factor of the coordination number $\frac{Nz}{2}$.² This mean field Hamiltonian describes a system of non-interacting spins in an *effective magnetic field* that depends on the magnetization m and the external field h .

The average magnetization per spin in the canonical ensamble can now be performed easily applying the mean field approximation:

$$\begin{aligned} \mathbb{E}_c \left[\frac{1}{N} \sum_i s_i \right] &= \sum_{m \in [-1, \dots, +1]} m \sum_{\{s_i\}} \delta \left(\sum_i s_i - Nm \right) \frac{e^{-\beta H_{mf}(m)}}{Z} \\ &= \frac{1}{Z} \sum_m m e^{-\beta H_{mf}(m)} \left[\sum_{\{s_i\}} \delta \left(\sum_i s_i - Nm \right) \right], \end{aligned}$$

where the last term identifies a sum over all the configurations of the spins that correspond to a given magnetization m . Note that a configuration of $N_\uparrow$ of up spins and $N_\downarrow$ of down spins corresponds to a magnetization

$$m = \frac{N_\uparrow - N_\downarrow}{N} = \frac{2N_\uparrow - N}{N} = 2\frac{N_\uparrow}{N} - 1.$$

The number of such configurations is given by the binomial coefficient:

$$\binom{N}{N_\uparrow} = \frac{N!}{N_\uparrow!(N - N_\uparrow)!} = \Omega(N_\uparrow),$$

which we can approximate using Stirling's formula for large N :

$$\log(\Omega(N_\uparrow)) \approx N \log N - N_\uparrow \log N_\uparrow - (N - N_\uparrow) \log(N - N_\uparrow),$$

which after some algebraic manipulations can be rewritten as

$$\log(\Omega(N_\uparrow)) \approx N \left[\log(2) - \frac{(1+m)}{2} \log(1+m) - \frac{(1-m)}{2} \log(1-m) \right].$$

Thus we have found an expression for the number of configurations that correspond to a given magnetization m , which can be used to compute the final form of the magnetization:

$$\mathbb{E}_c \left[\frac{1}{N} \sum_i s_i \right] = \frac{1}{Z} \sum_m m e^{-\beta N f(m)} = \frac{1}{Z} \sum_m m e^{-\beta H_{mf}(m)} \Omega(m),$$

where we have defined the function $f(m)$ as

$$f(m) = -hm - \frac{Jzm^2}{2} - T \left[\log(2) - \frac{(1+m)}{2} \log(1+m) - \frac{(1-m)}{2} \log(1-m) \right],$$

which can be interpreted as a *free energy density* (per site) that depends on the magnetization m . Indeed we have the last sum over the magnetization m getting dominated by the minimum of the function $f(m)$ in the thermodynamic limit $N \rightarrow \infty$, while the other terms get suppressed by the factor $e^{-\beta N f(m)}$. Thus the average magnetization can be approximated as

$$\mathbb{E}_c \left[\frac{1}{N} \sum_i s_i \right] \approx m^*, \quad \left. \frac{\partial f(m)}{\partial m} \right|_{m^*} = 0,$$

where m^* is the value of m that minimizes the function $f(m)$.

²When we count the same site for each of its n.n., we are counting each s_i a number of z times, but we have to divide by 2 to avoid double-counting. Thus the total is $N \cdot \frac{z}{2}$.

2.2.3 | Landau-Ginzburg Effective Field Theory

Appendices

A | Notation and Conventions

B | Computations and further developments